Ramadhan Adam Zome

Machine Learning Engineer · ML for Security · Representation Learning

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EDUCATION

Pan African University Institute for Basic Sciences, Technology and Innovation (PAUSTI) 2025 – 2027 (expected)
M.Sc. Artificial Intelligence and Machine Learning · Pan African University scholarship Nairobi, Kenya

Hiroshima University, Dependable Systems Laboratory Aug 2026 – Jan 2027
Special Research Student: training the thesis's byte-level encoder on the lab's GPUs Higashi-Hiroshima, Japan

Kenyatta University 2019 – 2023
B.Sc. Statistics and Programming · First Class Honours · county government scholarship Nairobi, Kenya

EXPERIENCE

Graduate Researcher, Open-World Malware Detection (M.Sc. thesis) 2025 – present
PAUSTI and Hiroshima University Python, PyTorch, LIEF, HDBSCAN, AWS S3

- Built and pretrained a byte-level transformer that reads Windows executables by file region, with a learned structural attention bias; self-supervised on **9.9M malware binaries** (208K steps, 140 A100-hours).
- Engineered fault-tolerant ingestion of 9.9M files from S3 into 1,986 memory-mapped shards: process-isolated parsing, memory ceilings, atomic writes; peak **115K files/hour**, under 0.7% retried or skipped.
- Detected malware families never seen in training with **0.81 AUROC** (target 0.70), using multi-prototype families, regularized Mahalanobis distance and per-family thresholds.
- Showed with controlled baselines that pretraining adds signal: 0.80 family recall on 102 families, ahead of no-training references on 10 of 10 splits. Presenting the work in Hiroshima on 17 Oct 2026.

Innovative Toll Solutions Dec 2023 – Aug 2025
Software Engineer Nairobi, Kenya

- Cleaned, explored and modelled operational logs and transaction data; built dashboards and reporting pipelines used by operations teams.
- Ran log-level anomaly detection, alert triage and traffic-pattern analysis; supported pipeline resilience work after DDoS incidents.

PROJECTS

Production ML on Kubernetes | FastAPI, k3d, GKE, Helm, GitHub Actions github

- Deployed a prediction API to GKE through GitHub Actions CI/CD, with a PostgreSQL StatefulSet for prediction logs, SOPS-encrypted secrets, Gateway API routing and Prometheus, Grafana and Loki monitoring.

Explainable Deepfake Detection with RAG | PyTorch, FAISS, FastAPI, React live demo

- Trained an Xception classifier on 140K real and GAN faces (about 97% held-out accuracy on 20K images) and added retrieval-augmented explanations, built without LangChain; FastAPI on Hugging Face Spaces, React on Vercel.

Phish-Transformer | PyTorch, Flask, Chrome MV3 github

- A 45K-parameter character-level transformer for phishing URLs (test AUC about 0.98) that runs on CPU behind a Flask API and warns users through a Chrome extension.

Windows PE Internals: sco-pe and raw-pe | C, NASM x86-64 github

- Wrote a PE parser in C with defensive bounds checks and unit tests (headers, sections, imports, exports, overlays), and hand-assembled a working 2,560-byte Win64 executable in NASM with no linker.

Swahili GPT | PyTorch, A100 model

- Trained a character-level GPT from scratch on 65M characters of Swahili news; used it to evaluate an A100 cloud platform for PAUSTI and wrote the decision report.

SKILLS

Languages	Python, C, SQL, Bash, x86-64 assembly (NASM), JavaScript
ML	PyTorch, TensorFlow, scikit-learn, Hugging Face, FAISS, HDBSCAN, contrastive and self-supervised learning
Systems	Docker, Kubernetes, Helm, GitHub Actions, GKE, FastAPI, PostgreSQL, Prometheus, Grafana, Linux
Security	PE format, static and dynamic malware analysis, reverse engineering, REMnux, FLARE-VM
Writing	Technical articles on PE internals, GPU memory and transformers: medium.com/@ramadhanzome4